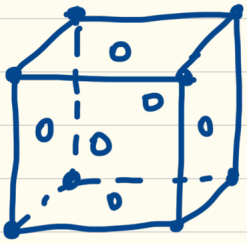
Homework 1

1.1 1.6 1.10 两道补充题

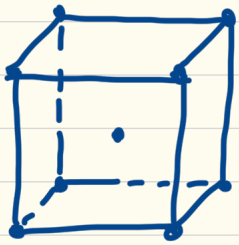
1.1 试求单位元胞中原子数：(a)面心立方体；(b)体心立方体；(c)金刚石晶格。

解：(a)



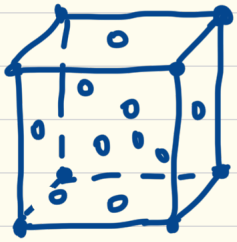
$N = 8\times\frac{1}{8}+6\times\frac{1}{2}=4$

(b)



$N = 8\times\frac{1}{8}+1=2$

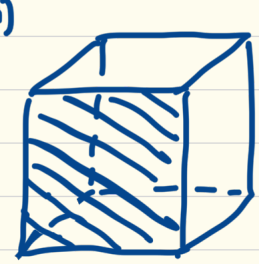
(c)



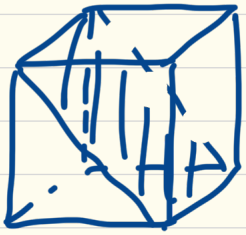
$N = 8\times\frac{1}{8}+6\times\frac{1}{2}+4=8$

1.6 画出简单立方体的下列晶面：(a)(100)；(b)(011)；(c)(111)；(d)(432)。

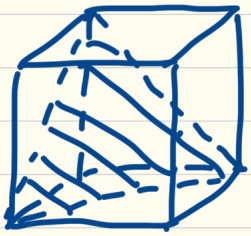
(a)



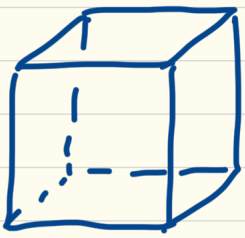
(b)

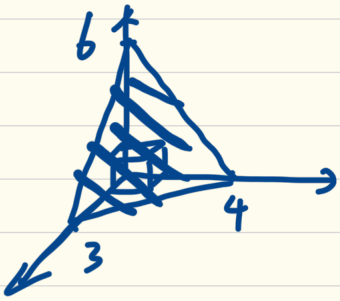


(c)



(d)

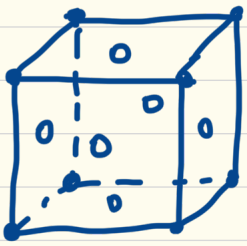


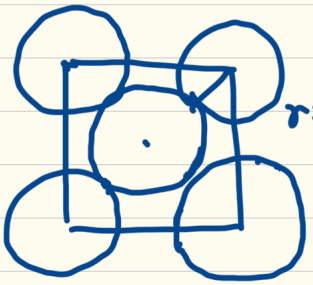


$\frac{1}{4}:\frac{1}{3}:\frac{1}{2}=3:4:6$

1.10 考虑一个面心立方体。设原子是硬球，且相邻两球的表面接触。已知原子半径为1.34 Å：(a)计算晶体内原子体密度；(b)计算最近(110)面之间距；(c)计算(100)面内原子面密度(cm^{-2})。

解.





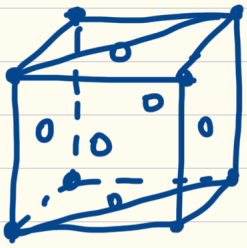
$1Å=1\times10^{-8}cm$

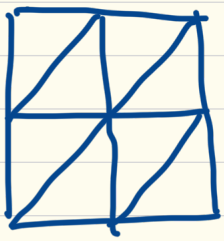
(a) 元胞内原子数 $=8\times\frac{1}{8}+6\times\frac{1}{2}=4$

元胞体积 $=(2\sqrt{2}r)^3=5.44\times10^{-23}cm^3$

体密度 $=\frac{\text{元胞内原子数}}{\text{元胞体积}}=7.35\times10^{22}$ 原子/cm^3

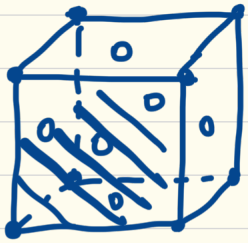
(b)

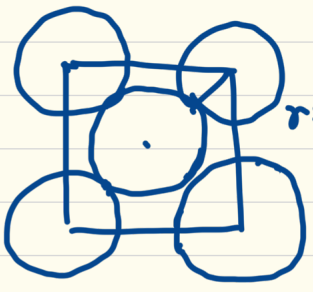




$d=2r=2.68\times10^{-8}cm$

(c)

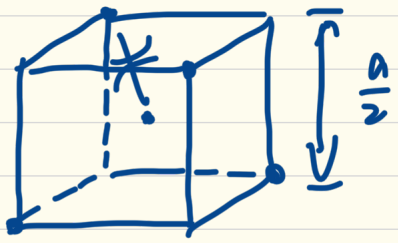




面密度 $=\frac{\text{晶格平面原子数}}{\text{晶格平面表面积}}=\frac{2}{(2\sqrt{2}r)^2}=1.39\times10^{15}$ 原子/cm^2

补充题

1. 计算等径圆球做金刚石最密堆积的空隙率。【3位有效数字】



解：$4r=\frac{\sqrt{3}}{2}a \Rightarrow r=\frac{\sqrt{3}}{8}a$

$V_{原子}=8\times\frac{4}{3}\pi r^3=\frac{32}{3}\pi r^3$

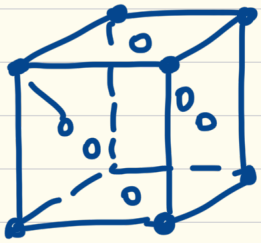
$V_{晶胞}=a^3$

$空隙率=1-\frac{V_{原子}}{V_{晶胞}}=1-\frac{\frac{32}{3}\pi(\frac{\sqrt{3}}{8}a)^3}{a^3}$

$=1-\frac{\sqrt{3}}{16}\pi\approx 0.660$

2. 面心立方紧密堆积的金属晶体，其原子量为M，密度是8.94 g/cm3。试计算器晶格常数和最小原子间距。【注意有效数字】

晶格常数 a　　最小原子间距 d



M：g/mol

N_A：$6.022\times10^{23}\ mol^{-1}$

解：$n=8\times\frac{1}{8}+6\times\frac{1}{2}=4$

$\rho=\frac{n\times M}{a^3\times N_A}$

$\Rightarrow a=\sqrt[3]{\frac{n\times M}{\rho N_A}}=9.06\times10^{-9}\sqrt[3]{M}\ cm$

$d=\frac{a}{\sqrt{2}}=6.40\times10^{-9}\sqrt[3]{M}\ cm$